

Yohannes Wubshet Abate

Electrical Engineering Technology Student | Research & Academic Track

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Academic Profile

High-achieving Electrical Engineering Technology student with demonstrated ability to independently conduct publishable research alongside full-time coursework. Research interests include renewable energy systems, AI-enabled embedded devices, computational algorithms, and semiconductor physics. Author of four IEEE-accepted conference papers completed as extracurricular research initiatives. Passionate about applying fundamental physics principles to breakthrough engineering solutions.

Education

Higher Colleges of Technology (HCT)

Bachelor of Electrical Engineering Technology | 2023 – Present

- Full merit-based national scholarship awarded through national entrance exam performance
- Ranked among top 273 students nationally in Grade 12 leaving examination
- Consistently achieved excellent academic standing over 2.5 consecutive years
- Strong performance across core engineering courses including Electronics, Control Systems, Communication Systems, and Industrial Automation

Relevant Coursework: Electronics I & II, Signals and Systems, Control Systems, Communication Systems, Renewable Energy Systems, Industrial Automation, Programming with Python & MATLAB

Bahir Dar STEM Incubation Center (Affiliated with Bahir Dar University)

STEM Track Student | 2019 – 2022

- Selected through competitive regional admission process
- Completed university-level STEM training taught by Bahir Dar University instructors
- Gained early hands-on experience in programming and applied problem-solving (C#, C++, Python)

Publications

(IEEE-Accepted, pending publication 2026. First author on all publications)

1. AI-Powered Smart Walking Stick Using Multi-Sensor Fusion and Embedded Vision for Assistive Navigation – ICERAI 2026, IEEE
2. Development of Building-Integrated Photovoltaics: A Review on Performance Optimization and Techno-Economic Aspects – PEII 2025 (ASET), IEEE
3. Advances in Bifacial Photovoltaic Technology: A Short Review from 2000 to 2025 – PEII 2025 (ASET), IEEE
4. Improving the Statistical Quality and Security of Pseudo-Random Number Generators Using an Adder Accumulator – AEE 2025 (ASET), IEEE

Research Experience

Renewable Energy & Semiconductor Systems Research

Conducted analytical studies on bifacial photovoltaic technologies with emphasis on degradation mechanisms, investigating efficiency limitations related to material interfaces, carrier recombination, and structural factors. Performed systematic literature reviews spanning 2000–2025, analyzing efficiency trends and techno-economic feasibility of building-integrated photovoltaics to assess practical deployment scenarios.

AI-Based Assistive Navigation System

Designed AI-enabled embedded systems integrating ultrasonic and laser sensors with YOLO-based object detection on Raspberry Pi 5. Implemented sensor fusion and decision-making logic using Python, conducting experimental validation and performance analysis to optimize system reliability for assistive navigation applications.

Algorithmic Security & PRNG Research

Developed enhancement models for pseudo-random number generators using adder-accumulator structures, implementing and benchmarking algorithms in C++ including LCG and Mersenne Twister. Evaluated statistical robustness using PractRand and TestU01 test suites while analyzing entropy behavior and randomness quality from a mathematical perspective.

Ongoing Research Projects

Currently developing MATLAB/Simulink-based simulations of photovoltaic performance under dust accumulation and partial shading conditions. Also pursuing image restoration research using Generative Adversarial Networks (GANs) to improve reconstruction quality of blurred images.

Professional Experience

Artificial Intelligence Intern | CodSoft

Virtual Internship | February 2025 – March 2025

- Contributed to applied AI tasks and model implementation projects
- Demonstrated strong analytical and problem-solving skills across technical projects
- Received strong performance remarks for contributions to technical team initiatives

Certifications & Professional Development

Artificial Intelligence & Machine Learning

- AI & Machine Learning Mastery – UAE Artificial Intelligence Academy (Microsoft & Zaka, 2025)
- Linear Algebra for Machine Learning and Data Science – DeepLearning.AI (Coursera, 2025)
- Google Boost Your Productivity with AI – INJAZ UAE Partnership (2024-2025)

Python Programming Specialization – University of Michigan (Coursera)

- Python Basics (2025)
- Python Functions, Files, and Dictionaries (2025)
- Data Collection and Processing with Python (2025)
- Python Classes and Inheritance (2025)

Additional Technical Training

- Programming Fundamentals – Udacity (2025)

Hackathons & Competitions

- AI for Good Hackathon – Powered by Nokia (NYUAD Slush'D 2025)
- Flag Rush CTF Competition – Exploit3rs (December 2025)
- Sophomore Engineering Design Project Competition – HCT Dubai Campus (November 2025)
- 2nd Place – HCT Mascot Design Competition (2025)

Technical Skills

Programming & Scripting: Python (Intermediate), MATLAB (Intermediate), C++ (Intermediate), C#

AI & Machine Learning: Machine Learning fundamentals, model implementation, data processing, TensorFlow basics, GANs

Embedded Systems & Hardware: Raspberry Pi 5, Arduino, microcontrollers, ultrasonic & laser sensors, PCB design & manufacturing

Engineering & Simulation Tools: MATLAB/Simulink, Multisim, EasyEDA, Jupyter Notebooks, Google Colab

Research & Analysis: PractRand, TestU01, statistical analysis, technical writing, literature review methodology

Research Interests

- Semiconductor Physics & Device Behavior
- Photovoltaic Technologies & Renewable Energy Systems
- Nanoscale Energy Systems
- AI-Enabled Embedded Intelligence
- Computational Modeling of Physical Systems
- Algorithmic Security & Cryptography
- Quantum Computing (Long-term Goal)

Teaching & Academic Service

Peer Tutor – Higher Colleges of Technology

- Assisted students with core engineering and technical subjects
- Strengthened conceptual understanding through structured tutoring sessions
- Developed effective communication strategies for complex technical concepts

Leadership, Volunteering & Fellowships

Millennium Fellowship – 1Change 4All

United Nations Academic Impact & Millennium Campus Network (MCN)

- Selected for global leadership fellowship focused on advancing UN Sustainable Development Goals
- Engaged in community impact initiatives aligned with sustainability and social development

Community & Environmental Volunteering

- 153+ verified volunteer hours including sustainability initiatives and community projects
- Active participant in environmental awareness and social impact programs

Honors & Awards

- Paper Presenter/Author Recognition – ASET 2025 Conference (Power, Energy, and Industrial Innovations)
- 2nd Place – HCT Mascot Design Competition (2025)
- Merit-Based National Scholarship – Full tuition coverage, HCT (2023-Present)
- Top 273 National Ranking – Grade 12 Leaving Examination, Ethiopia

Languages

- **English:** Professional proficiency (spoken and written)
- **Amharic:** Native speaker

Professional Competencies

Research Methodology • Technical Writing • Critical Thinking • Problem Solving • Teamwork & Collaboration • Leadership • Project Management • Data Analysis • Independent Research • Academic Integrity